

Wireless Security Threats and Defenses — Guided Notes

ANSWER KEY

Unit 4.3.1 | CompTIA Network+ N10-009 Obj. 4.4 | N10-008 4.4

For instructor use / distribute after students complete the notes.

Question 1

WEP is broken because its 24-bit _____ repeats on busy networks within hours. Once an attacker collects enough repeated IVs, they can mathematically recover the _____. The correct action when WEP is found in production is to _____.

Answer: Initialization vector (IV); encryption key. Replace it immediately — WEP provides no effective security and cannot be fixed through configuration.

Question 2

WPA replaced WEP's broken encryption with _____ (TKIP), which was designed to run on existing hardware. WPA2 replaced TKIP with _____ using AES encryption. Why was TKIP considered a stopgap rather than a long-term solution?

Answer: TKIP (Temporal Key Integrity Protocol); CCMP. TKIP was built around the RC4 cipher WEP used, inheriting its fundamental weaknesses — it was designed for backward compatibility, not long-term security.

Question 3

WPA2-Personal uses a _____ shared by all devices. WPA2-Enterprise uses individual credentials authenticated via a _____ server. Why does Personal mode fail in organizational environments?

Answer: Pre-shared key (PSK/passphrase); RADIUS. A shared PSK cannot be revoked for a single user — changing it requires reconfiguring every device, and compromise by one user affects all.

Question 4

WPA3-Personal replaces PSK with _____ (SAE), which provides _____ secrecy — meaning previously captured traffic cannot be decrypted even if the passphrase is later compromised.

Answer: Simultaneous Authentication of Equals; forward.

Question 5

802.11w Protected Management Frames (PMF) is mandatory in WPA3. It encrypts and authenticates _____ frames, which defeats _____ attacks that force clients off a legitimate AP.

Answer: Management (deauthentication/disassociation); deauthentication (deauth) attacks.

Question 6

A WIDS passively _____ the RF environment for rogue APs and unauthorized activity. A WIPS adds the ability to actively _____ rogue devices by sending deauthentication frames to clients connected to them.

Answer: Monitors; contain/suppress.

Question 7

A captive portal intercepts _____ traffic and redirects unauthenticated clients to a login page. Name two security weaknesses of captive portals that make them unsuitable as a primary security control.

Answer: HTTP (unencrypted); weaknesses include: MAC address spoofing bypasses authentication, pre-authentication traffic is unencrypted on open SSIDs, and the portal only controls access — it does not encrypt ongoing sessions.

Question 8 — Real World Application

REAL WORLD: A school deploys WPA2-Personal for its staff SSID. Within a week, the passphrase appears on a student group chat. IT changes the password, and it leaks again within three days. Identify the root architectural problem and describe the correct solution.

Answer: Root problem: WPA2-Personal uses a single shared secret that cannot be revoked per user — any one person with the password can share it. Solution: migrate to WPA2/WPA3-Enterprise with 802.1X and RADIUS authentication tied to Active Directory. Each user authenticates with their own credentials, which can be revoked individually without affecting other users.